

14 Drag & Aero Characterisation

The drag of Stick Bug is crucial in competitive range and fuel efficiency performance. For a transport aircraft flying in the transonic range at cruise, parasitic and induced drag are major contributors to the overall drag of the aircraft. These components and other notable ones are discussed in this section, covering their effect and methods of calculation.

14.1 Parasitic Drag

The dominant contribution to parasitic drag in high Reynolds number flows is boundary layer drag. As such a boundary layer component buildup method was employed that uses flat plate assumptions to determine the parasitic drag of the entire aircraft throughout its regimes of flight (Torenbeek, 1976) by applying Schlichting's equation for incompressible flow with a compressibility correction factor as seen in section 17.1. The state of the boundary layer was assumed to be turbulent over Reynolds numbers of 5×10^7 , and an additional 15% was added to results to account for a factor of safety. This was done within MATLAB and took wetted area values from CAD models that are noted to more accurately represent fuselage area compared to the fuselage utilised in the following technique.

To compare with these values NASA's open-source parametric aircraft geometry tool OpenVSP (NASA, 2025) was used to calculate parasitic drag in parallel using the Schlichting Compressible method as well as the Blasius model for laminar flow sections. Both methods are mathematically detailed and reported in section 17.1. The components of each aircraft component for both methods and the overall sum can be seen summarised in Table 14-1.

Table 14-1: Parasitic Drag Components at Cruise

Component	MATLAB	Open VSP	Average
Wings	0.0062	0.0050	0.0056
Horizontal Tail	0.0021	0.0009	0.0015
Vertical Tail	0.0006	0.0004	0.0005
Fuselage	0.0075	0.0057	0.0066
Engine Nacelles	0.0052	0.0015	0.0034
Strut	0.0007	0.0033	0.0020
Totals	0.0223 (incl. excrescence)	0.01690 (+ 15% excrescence)	0.0196

Given the factors of both buildup methods, a mean will be taken as the most representative parasitic drag value of 0.0196.

14.1.1 Parasitic Drag Variance

Parasitic drag varies with flight speed and altitude in line with variance in the Reynolds number that characterises the stream over each component of the plane. The key areas are take-off, landing, and cruise. The parasitic drag in these regimes is critical as it informs the total drag of the airplane in these conditions which can be used to estimate range. Using the initial method within MATLAB (MATLAB, 2025) the variance of these formulas through a wide range of flight regimes can be seen in Figure 14-1.

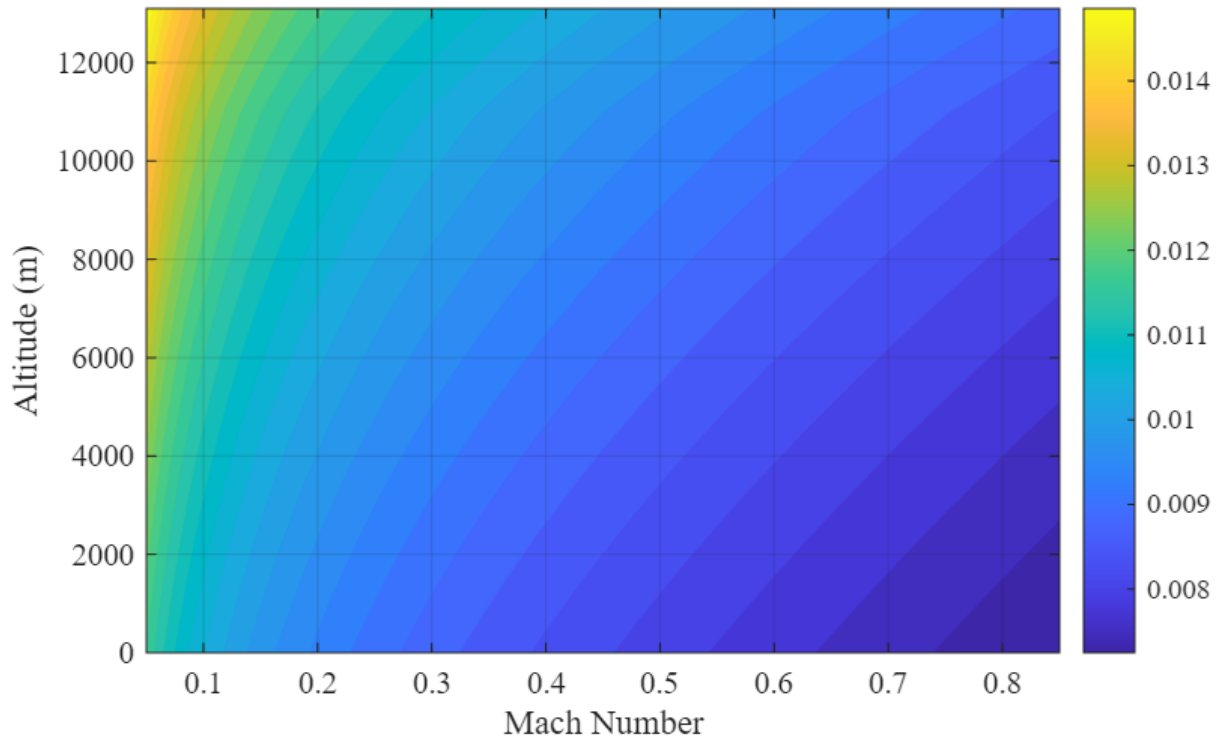


Figure 14-1: Parasitic Drag Coefficient Variation

This data becomes relevant when calculating the total drag in the following section. To account for the parasitic drag change during flight stages such as take-off and landing, when landing gear and other components can increase the parasitic drag, an empirical estimation method was used.

14.2 Total Drag

Now that the parasitic drag has been calculated the total drag can be calculated for the plane during different flight regimes. Specifically, the induced drag from lift generated by the main wing, horizontal stabiliser and pressure drag from the fuselage will be analysed. This will be analysed in the same way as parasitic drag, looking into both theoretical and OpenVSP estimates.

The equation of drag for any lifting body involves a parasitic drag component and an induced drag component shown in equation 14-1.

$$C_D = C_{D,0} + \frac{C_L^2}{\pi A R e}$$

Equation 14-1: Total Drag Formula

The previously calculated parasitic drag coefficient is a sum of all airplane components; the induced drag will be found in the same manner. For each lifting component, the calculated efficiency factor e , the coefficient of lift, and the aspect ratio of each lifting body will be used to calculate the 3D wing drag coefficient. This covers the main wing, and the horizontal tail.

For the fuselage, the main drag contribution is the skin friction drag, in addition to this there is base drag caused by the low-pressure region behind the tail of the fuselage. The airflow will be assumed to be attached to the fuselage until the base (flat tail section). This can be approximated by equation 16-2.

$$C_{D_{base}} = \left\{ \frac{0.029 \left(\frac{d_b}{d} \right)^3}{\sqrt{C_{D_{sf}} \left(\frac{S_{ref}}{S_{wet}} \right)}} \right\} \left(\frac{S_{wet}}{S_{ref}} \right)$$

Equation 14-2: Base Drag Estimation

Using induced drag for all lifting surfaces and fuselage base drag in addition to parasitic drag for a range of conditions results in the following values summarised in Table 14-2 is further detailed in section 17.1.

Table 14-2: Total Drag Values

Case	Drag Coefficient (C_D)
Clean	0.0361
Take-off	0.1321
Landing	0.2345

14.3 Wave Drag

Due to the supercritical aerofoils being used Stick Bug can fly at a Mach number of 0.85 without suffering large shock wave drag penalties over the lifting surfaces. The fuselage will have accelerated flow to a lesser extent and therefore be neglected.

14.4 Drag Compared to Lift

For the clean, landing and take-off configurations max L/D is known through the performance of the wing at a specific angle of attack, and the addition of high-lift devices as previously mentioned. This can be compared to total aircraft drag for three flight regimes including the contribution of the horizontal tail in landing and take-off configurations. The results can be seen below in Table 14-3.

Table 14-3: Total Drag and Lift Summary

Configuration	C_L (Wing & Tail)	C_D (Entire Aircraft)	L/D
Cruise	0.8730	0.0361	24.1828
Take-off	2.6550	0.1321	20.0984
Landing	3.7394	0.2345	15.9463

14.5 OpenVSP Simulations

Reference parasitic drag estimates and the neutral point of the aircraft was obtained through OpenVSP. This allowed verification of analytical parasitic drag results and neutral point calculations. As seen in Table 14-1 results from both methods were in strong agreement. To model the aircraft in OpenVSP the main components were created to match the final configuration of the aircraft. This layout can be seen in Figure 14-2: OpenVSP Model.

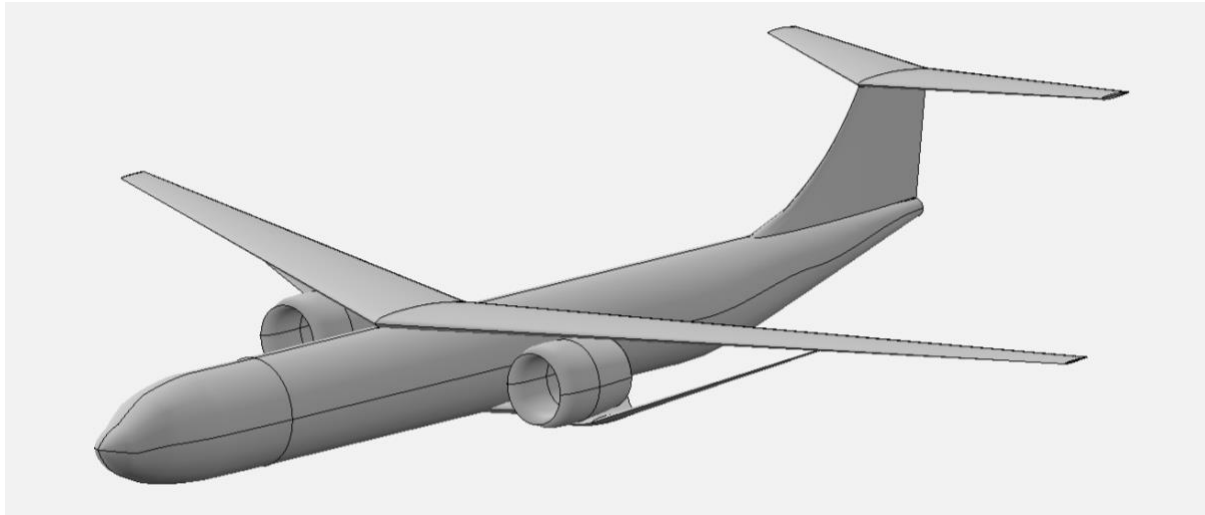


Figure 14-2: OpenVSP Model

With this, a steady stability analysis was conducted at cruise which resulted in a neutral point location of 30.1 m from the nose of the aircraft. This value was used to inform the static margin of the aircraft in this configuration.

14.6 Wing Flutter Considerations

High aspect ratio (AR) wings come with the benefits of lower induced drag which is a large hindrance to transport jet performance. The drawback of this is with slender structures indicative of high AR wings, comes structural problems. Wings become less stiff and are more susceptible to bending and flutter effects as opposed to lower AR, more ‘stubby’ wings. Wing flutter is a dynamic phenomenon where a wings inertial, aerodynamic and structural loads combine to show oscillatory behaviour, most commonly the flexion of the wing tips. This is a highly complex aeroelastic phenomenon that can result in diverging oscillatory behaviour which can increase the amplitude of wing tip deflection beyond structural limits.

Due to the complexity of the wing strut interaction on aeroelasticity this phenomenon could not be fully characterised within the time constraints of the project. As such a similar concept aircraft, NASA’s X-66 (NASA, 2015) will be used to infer flutter characteristics for an approximate benchmark.

The NASA study considered a dive speed of Mach 0.82 with a 1.15 FOS required by FAA regulations. A study of the wing with and without fuel representative of MTOW was conducted and found that for all dive speeds below Mach 1 the critical flutter speed (speed at which flutter begins to diverge) exceeded the FAA requirements, with no indicative divergence behaviour being observed at Mach 0.9. It was also found that the additional weight penalties from bolstered structures to resist these flutter effects was lesser than that of a cantilevered wing, further reinforcing the relevance of the SWB. With this information the maximum dive speed of Stick Bug would likely be above 0.85, however this number can only be speculative without a dedicated flutter and aeroelastic study.

Additionally, flutter suppression control systems can be used to further reduce this flutter effect. This would involve deflecting the outboard ailerons to suppress growing oscillatory modes (Townsend & V. Dogett, Jr.).

17 Aerodynamic Characterisation

17.1 Parasitic Drag Estimation

Torenbeek Parasitic Drag Buildup

The first method to calculate parasitic drag was using a component buildup method detailed by Torenbeek (Torenbeek, 1976) which utilises Schlichting's equation for turbulent incompressible flow with a correction factor applied. This method uses the Reynolds number to estimate the parasitic drag component of aircraft structures based on their boundary layer drag. First the Reynolds number using the reference length of a given component is used.

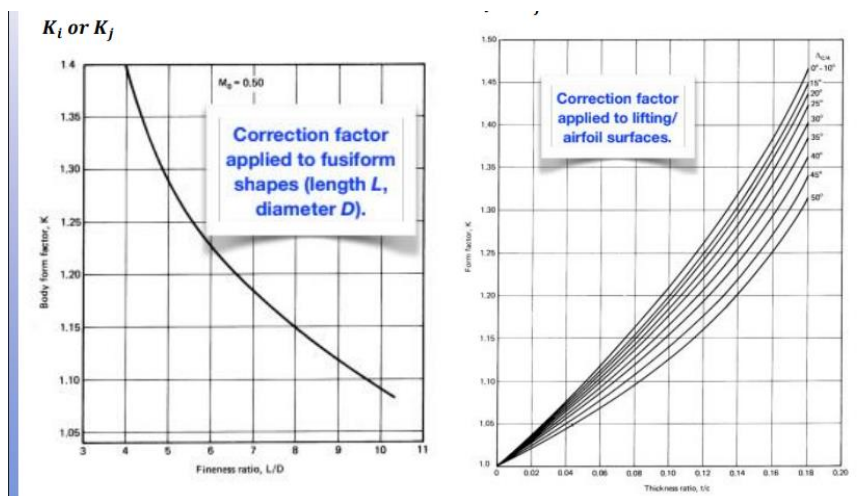
$$Re = \frac{\rho V l}{\mu}$$

For lifting surfaces the reference length is the mean aerodynamic chord. The total parasitic drag coefficient for components 1 through n can be summarised by

$$C_{d,0} = 1.15 \frac{1}{S_w} \left[\sum_{i=1}^n K_i C_{fi} S_i \right]$$

Where S_w is the wetted area of the main wings, K_i is a correction factor based on whether the object is a fusiform or lifting body type shape, and finally S_i is the wetted area of the component. The additional 1.15 factor is an increase to account for excrescence drag.

The correction factor K is determined from empirical graphs sourced from the flight vehicle design course notes (Monash University, 2025).



Schlichting's equation can then be used to calculate frictional coefficients for each aircraft part.

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58} (1 + 0.144 M^2)^{0.65}}$$

It should be noted that normally there is a section of laminar flow before a turbulent boundary layer develops. This method does not account for that, only considering Reynolds number to dictate fully turbulent or fully laminar flow, hence for the critical flight regimes such as cruise, this method is expected to overestimate the parasitic drag value.

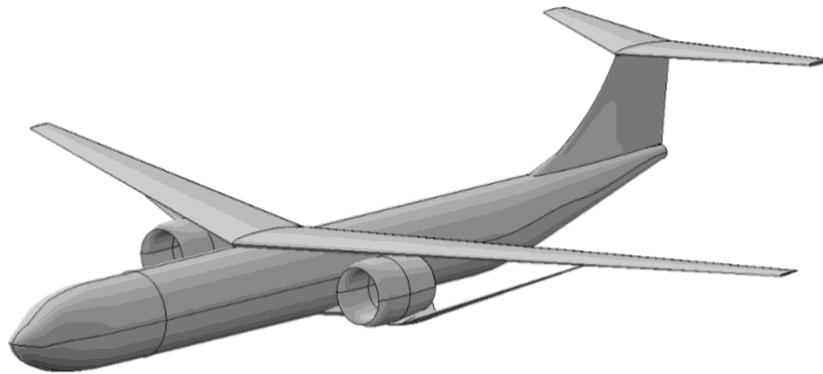
Open VSP Parasitic Drag Estimation

To verify and compare drag buildup methods, the software OpenVSP (NASA, 2025) was also used to calculate parasitic drag. OpenVSP allows for the parameterised construction of aircraft geometry that can then be analysed using various modules and techniques. The parasitic drag tool was used to calculate the total parasitic drag of the aircraft using a slightly modified Schlichting compressible method for turbulent flow calculations and the Blasius equation for laminar flow calculations, both detailed below.

$$C_{f_turb} = \frac{0.455}{\log(Re)^{2.58}}$$

$$C_{f_lam} = \frac{1.32824}{\sqrt{Re}}$$

The planes major surfaces were constructed within OpenVSP and ran through the parasitic drag tool with these equations. The diagram of the component geometries can be seen below.



It should be noted that the simplifications of fuselage shape will result in a slightly lower wetted area, resulting in slightly lower parasitic drag contributions. The average between Open VSP and MATLAB calculated values will be taken as such to leverage the benefits of both conservative results and more complex analysis.

17.2 Analytical Total Drag Calculations

The equation for total drag of a lifting body is given as

$$C_D = C_{D,0} + \frac{C_L^2}{\pi A Re}$$

Simple component drag buildup can be used to obtain a drag coefficient for the aircraft at different regimes.

For the fuselage the base drag caused by the low pressure separated airflow at the tail can be accounted by the following equation

$$C_{D_{base}} = \left\{ \frac{0.029 \left(\frac{d_b}{d} \right)^3}{\sqrt{C_{D_{sf}} \left(\frac{S_{ref}}{S_{wet}} \right)}} \right\} \left(\frac{S_{wet}}{S_{ref}} \right)$$

Adding previously calculated parasitic drag onto the calculated induced drag reaches the following results



Case	Wings	Horizontal Tail	Fuselage
Clean	0.0224	0.0021	0.0078
Take-off	0.0682	0.0569	0.0085
Landing	0.1618	0.0487	0.0086

Summing parasitic and induced drag gives a total coefficient of drag for the aircraft in each of its configurations.

Case	Drag Coefficient (C_D)
Clean	0.0361
Take-off	0.1321
Landing	0.2345